What Kadra Does

Kadra is the programmable infrastructure layer for private real estate funds, replacing manual, siloed back-office workflows with a blockchain-based settlement engine. By anchoring legal documents on-chain and hardcoding compliance rules directly into an ERC-3643 security token, Kadra automates administration, eliminates wire fraud, and reduces settlement times from weeks to seconds while maintaining a single, mathematically verifiable system of record.

Technical Components

1. Core Architecture

Next.js Frontend: A clean, institutional-grade UI (Tailwind CSS) mimicking standard banking software.

Local Backend: A local database to store off-chain data (names, emails) and a local Hardhat node (via Scaffold-ETH 2) to manage instant on-chain transactions.

2. Fund Initialization

PDF Upload Engine: An interface to upload a dummy Master LPA PDF.

Hashing Engine: A function that automatically generates a SHA-256 cryptographic hash of the uploaded PDF and anchors it to the smart contract to permanently prove the document's state.

3. Identity Permissioning

Investor Registry UI: A dashboard to map real-world investor identities to specific wallet addresses.

Whitelist State: Logic to change an investor's status from "Unknown" to "Verified," explicitly authorizing their wallet on the blockchain to hold the fund's asset.

4. Settlement Validator

Wire Upload: An interface to upload mock bank wire receipts.

3-Way Match Logic: A backend script that checks if the uploaded receipt perfectly matches the Investor Identity, the Capital Amount, and the specific Deal Reference.

Trigger: The function that flips the state from "Pending" to "Settled" upon a successful match, triggering the smart contract to mint tokens.

5. Smart Contract Enforcement

ERC-3643 Deployment: The actual security token contract representing fund ownership.

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Transfer Restrictions: Hardcoded conditional transfer logic that intercepts any transfer attempt, checks the Identity Registry, and deterministically blocks transactions sent to unverified wallets.

6. Yield Management

Distribution Calculator: A module that calculates pro-rata distributions for each investor based directly on the real-time token cap table.

Execution Button: A trigger to simulate the instant payout of yield across all verified wallets.

7. Data Generation & Audit

Live Cap Table: A frontend component that dynamically reads token balances directly from the blockchain.

Audit Packet Generator: A one-click export function that bundles the original PDF, the cryptographic hash, and a CSV log of all blockchain transactions into a downloadable ZIP file.

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